MATH 205

# One-to-One, Onto & Inverses

LECTURE 15 · ONE-TO-ONE, ONTO & INVERSES

*Can you always undo a transformation? The answer hinges on two questions: does it lose information, and does it reach everywhere?*

SAJAL SOHAIL RANA

WELCOME BACK

# Where Today Fits

### Last time

- The formal definition, for any vector spaces V and W
- Proving linearity: matrices, derivatives, integrals
- Kernel and range — what's lost, and what's reached

### Today

- One-to-one: ker(T) = {0}, and a pivot-column shortcut
- Onto: R(T) = W, and a pivot-row shortcut
- Identity and zero transformations, formally
- Inverse linear transformations — when “undo” exists

### Course Roadmap

1. Systems of Linear Equations
2. Matrices & Matrix Algebra
3. Determinants
4. Vector Spaces
5. **Linear Transformations**

LECTURE 3 of 4

*Both shortcuts today are just row reduction in disguise — the exact skill you built back in Lectures 10 and 11.*

*By the end: given any matrix, you'll read off one-to-one, onto, and invertibility just from its pivots.*

7.8 ONE-TO-ONE

# Does T Lose Information?

*If two different inputs ever produce the same output, you can never tell them apart afterward.*

*One-to-one is the promise that this never happens.*

7.8 DEFINITION

# One-to-One

A linear transformation T : V → W is ONE-TO-ONE if T maps DISTINCT vectors in V into DISTINCT vectors in W.

$$T(\underline{u}) = T(\underline{v}) \implies \underline{u} = \underline{v}$$

First, a reminder of what T being a well-defined MAPPING already requires — one-to-one is a stronger ask on top of this:

**NOT a mapping — two images**



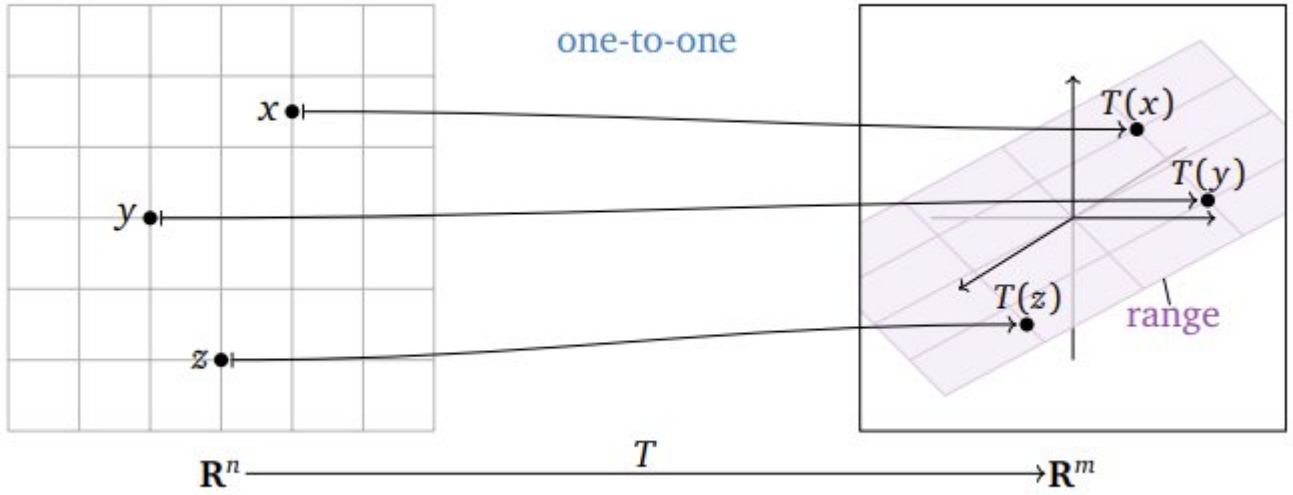
*$v_3$ has TWO images*

**NOT a mapping — missing image**



*$v_4$ has NO image*

# One-to-One ⇔ Trivial Kernel




$$T \text{ is one-to-one} \iff \ker(T) = \{\underline{0}\}$$

*This is the single most useful fact in today's lecture: checking "one-to-one" reduces to checking "kernel is trivial" — something you already know how to do.*

7.8 SHORTCUT

# For Matrix Transformations: Read the Pivots

For T(x) = Ax, the kernel test translates directly into row reduction:

$$T \text{ one-to-one} \iff \text{every column of } A \text{ has a pivot} \iff \text{columns of } A \text{ independent}$$

Why: ker(T) = {0} means Ax = 0 has ONLY the trivial solution — which happens exactly when there are no free variables, i.e. every column is a pivot column. No free variables also means the columns of A are linearly independent — the same fact, two vocabularies.

Pp 131 - 134 ▸ Try the live demo at gatech.edu/ila

WORKED EXAMPLE

# Checking One-to-One via Pivots

Let T(x) = Ax where:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Row-reduce: $R_3 \to R_3 - R_1$ gives (0,−2); then $R_3 \to R_3 + 2R_2$ gives (0,0). A pivot survives in BOTH columns — no free variables.

$$\text{Both columns have a pivot} \implies T \text{ is one-to-one}$$

$$3 \text{ rows},\ 2 \text{ columns} \ \ (m > n) \implies T \text{ is NOT onto}$$

*(We'll use that second fact in a moment — hold that thought.)*

QUICK REFERENCE

# A Shape Test You Can Do at a Glance

$$m < n \quad \text{(more columns than rows)} \implies T \text{ is NOT one-to-one}$$

Why: with more columns (n) than rows (m), row reduction can produce at most m pivots — leaving at least n−m columns without one. Those become free variables, so ker(T) always contains more than just 0. Imagine a 3 x 5 matrix ...

*Picture it: you're trying to fit more independent directions (n) into a smaller output space (m) than fits — something has to collide.*

7.9 ONTO

# Does T Reach Everywhere?

*One-to-one asked about collisions on the way in. Onto asks a completely different question: does every point in the codomain actually get hit?*

7.9 DEFINITION

# Onto

A linear transformation T : V → W is ONTO if the range of T is all of W — every vector in the codomain is T(v) for some v in V.

$$R(T) = W$$




*Every vector in W has at least one arrow pointing to it — no gaps, though some may share an input.*

7.9 SHORTCUT

# For Matrix Transformations: Rows This Time

The onto test has the same flavor, but checks rows instead of columns:

$$T \text{ onto} \iff \text{every row of } A \text{ has a pivot} \iff \text{columns of } A \text{ span } R^m$$

Why: R(T) = Col(A), and Col(A) fills all of $\mathbb{R}^m$ exactly when the columns of A span $\mathbb{R}^m$ — which happens exactly when row reduction leaves no zero rows, i.e. a pivot in every row.

Pp 136 - 138 ▸ Try the live demo at gatech.edu/ila

WORKED EXAMPLE

# Is Our Transformation Onto?

Same A from a few slides ago — 3 rows, 2 columns, both columns pivoted:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Only 2 pivots exist (one per column) — but there are 3 rows. Row 3 can never get a pivot of its own.

So T is NOT onto — exactly matching the shape test from two slides ago (3 rows > 2 columns). T is one-to-one, but its range is only a 2-dimensional slice of the 3-dimensional codomain.

*One-to-one and onto are INDEPENDENT properties — a transformation can have either, both, or neither.*

QUICK REFERENCE

# The Mirror-Image Shape Test

$$m > n \quad (\text{more rows than columns}) \implies T \text{ is NOT onto}$$

Why: row reduction can produce at most n pivots (one per column, at best) — so if there are more rows (m) than columns (n), at least m−n rows are guaranteed to end up with no pivot, hence no zero row.

| | |
|---|---|
| **Wide (n > m)** | *can be onto, never one-to-one* |
| **Tall (m > n)** | *can be one-to-one, never onto* |
| **Square (m = n)** | *one-to-one ⇔ onto ⇔ invertible* |

7.10 IDENTITY, ZERO & INVERSES

# The Two Extremes, and Undoing T

*Identity is the best possible transformation — both one-to-one and onto. Zero is the worst. In between, one-to-one is exactly what lets you build an inverse.*

# Identity and Zero, Formally

### IDENTITY

I(v) = v for every v in V. It's trivially one-to-one (distinct inputs stay distinct) and trivially onto (every w = I(w)). Best possible behavior.

**ker(I) = {0}** **R(I) = V**

### ZERO TRANSFORMATION

$$T(\underline{v}) = \underline{0} \quad \text{for every } \underline{v} \in V$$

Every vector maps to 0 — the most extreme information loss possible. NOT one-to-one (unless V = {0}), and only onto when W = {0} too.

**ker(0) = V** **R(0) = {0}**

7.10 DEFINITION

# Inverse Linear Transformation

If T : V → W is linear and ONE-TO-ONE, then the inverse linear transformation is defined by $T^{-1}$ : R(T) → V, which maps each w in R(T) back to the unique v in V with T(v) = w.

$$T^{-1} : R(T) \to V$$

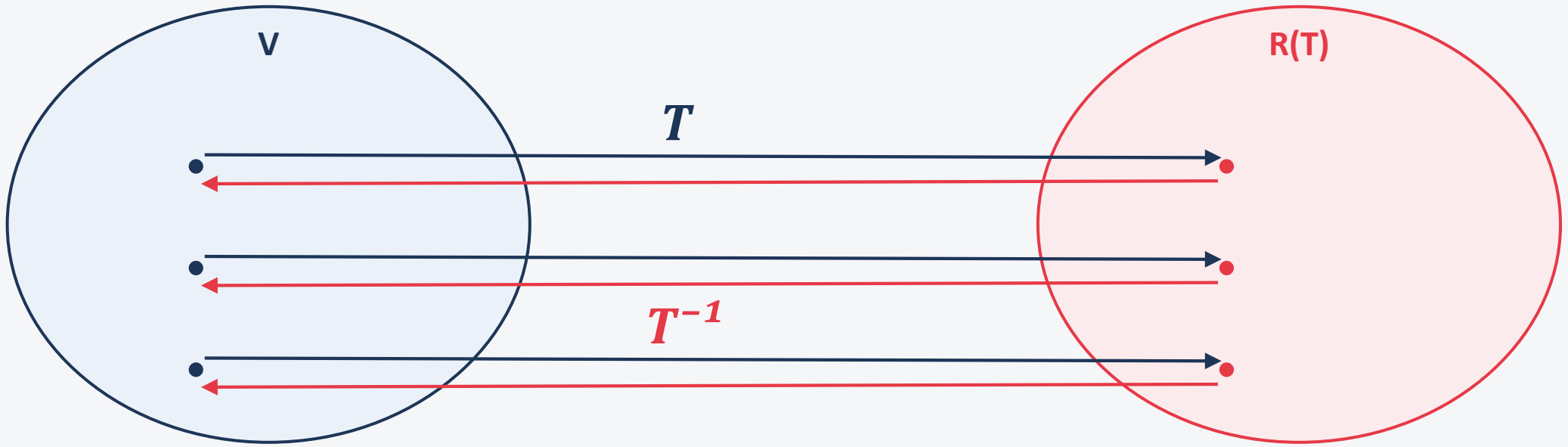
One-to-one is exactly what makes $T^{-1}$ well-defined: it guarantees each w in R(T) came from only ONE v, so “map it back” is unambiguous. $T^{-1}$ is defined on R(T) — which may or may not be all of W (that's the onto question, and it's a separate one).

*If T happens to be onto too, $T^{-1}$ is defined on the whole of W — T and $T^{-1}$ perfectly undo each other.*

PICTURE

# The Round Trip

T takes you from V into R(T). $T^{-1}$ takes you right back — always landing on the same starting vector.




*Every vector in R(T) has a well-defined path back — that's what one-to-one buys you.*

7.10 RESULT

# Inverting a Matrix Transformation

If $T : \mathbb{R}^n \to \mathbb{R}^n$ is multiplication by an invertible matrix A, the inverse $T^{-1} : \mathbb{R}^n \to \mathbb{R}^n$ is multiplication by $A^{-1}$.

$$T(\underline{x}) = A\underline{x} \text{ invertible} \iff A \text{ invertible}, \qquad T^{-1}(\underline{x}) = A^{-1}\underline{x}$$

This is the square case from slide 13: for $T : \mathbb{R}^n \to \mathbb{R}^n$, one-to-one, onto, and "A is invertible" are all the SAME condition. That's the Master Theorem territory — dozens of equivalent ways to say "A behaves as well as possible," many of which you've already met: det(A) ≠ 0, rank(A) = n, columns independent, columns span $\mathbb{R}^n$.

WORKED EXAMPLE

# Undoing a Rotation

Recall the rotation transformation from Lecture 13:

$$T : \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Rotating by −θ undoes rotating by θ — and swapping the sign of θ in cos and sin gives exactly the inverse matrix:

$$T^{-1} : \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \quad (\text{rotate by } -\theta)$$

*Geometrically obvious, algebraically confirmed: rotation matrices are always invertible (det = $\cos^2\theta+\sin^2\theta = 1 \neq 0$), and the inverse is just "rotate the other way."*

APPLICATION

# Undo Buttons, Everywhere

Any time a system needs to reverse a transformation exactly, it's relying on $T^{-1}$ existing.

| | |
|---|---|
| **Computer graphics** | undoing a rotation/scale to convert screen clicks back into world coordinates. |
| **Cryptography** | many ciphers are literally invertible linear maps — encoding is T, decoding is $T^{-1}$. |
| **Error-correcting codes** | recovering the original message requires the encoding map to be one-to-one. |
| **Robotics (forward/inverse kinematics)** | going from joint angles to hand position and back — though as we saw, that map often ISN'T linear, which is exactly why it's hard. |

*Every one of these starts with the same question we asked today: is the kernel trivial?*

RECAP

# Key Takeaways

### One-to-One

ker(T) = {0} — for matrices, a pivot in every column.

### Onto

R(T) = W — for matrices, a pivot in every row.

### Shape Tests

wide: can't be 1-1. tall: can't be onto. square: all-or-nothing.

### Identity / Zero

the two extremes — always both, or (almost) never either.

### Inverse Transformation

exists on R(T) exactly when T is one-to-one.

### Matrix Inverses

T(x)=Ax invertible ⇔ A invertible ⇔ A square, 1-1, and onto.

NEXT TIME

# Looking Ahead

**Before Next Lecture**

**Classify a transformation**

For $T : \mathbb{R}^3 \to \mathbb{R}^2$, $T(x,y,z)=(x+y, y+z)$ from last lecture's homework: is it one-to-one? Onto? Does $T^{-1}$ exist?

HOMEWORK 11

Lecture 16

## The Matrix of a Linear Transformation

- The standard-basis trick: a matrix's columns are $T(e_1)$, $T(e_2)$, …
- Building the matrix for a transformation you're only given a formula for.
- Closing worked applications tying the whole unit together.

*Reference: Anton & Rorres, §8.4 · ILA §3.3*

# Questions?

*One-to-One, Onto & Inverses · Lecture 15 complete*

**NEXT: THE MATRIX OF A LINEAR TRANSFORMATION**